LeanCode

Programming in Flutter

Animations

What are animations?

It is the motion on the screen - many still images shown in quick succession

Why animations?

Functional

- Meaningful transitions
- Provide visual feedback
- Show system status
- Help user get started (e.g. by displaying animation of desired interaction)

Delightful

- To demonstrate polish of the app
- To bring personality and humanity to the app
- To entertain users

Implicit vs Explicit Animations

Implicit Animations

Implicit animations

- AnimatedContainer
- AnimatedAlign
- AnimatedPositioned
- AnimatedPositionedDirectional
- AnimatedOpacity
- AnimatedPadding
- AnimatedTheme
- AnimatedCrossFade

And many more… See here: https://docs.flutter.dev/ui/widgets/animation

Implicit Animations

Do not require a controller!

```dart
class _PageState extends State<Page> {
  double _opacity = 1.0;

  @override
  Widget build(BuildContext context) {
    return AnimatedOpacity(
       opacity: _opacity,
       duration:  const Duration(miliseconds: 300),
       child: const Text('This will animate'),
    );
  }
}
```

Explicit Animations

AnimationController

We control the ~~traffic~~ animation

AnimationController

We are responsible for initializing it, but also for the disposal.

```dart
class _PageState extends State<Page> with SingleTickerProviderStateMixin{
  late final AnimationController _animationController;

  @override
  void initState() {
    super.initState();
    _animationController = AnimationController(vsync: this);
  }

  @override
  void dispose() {
    _animationController.dispose();
    super.dispose();
  }


  @override
  Widget build(BuildContext context) {
    return ...;
  }
}
```

AnimationController

With great responsibility,
comes great power.

```
_controller.isAnimating;
_controller.lastElapsedDuration;
_controller.status;
_controller.value;
_controller.velocity;
_controller.view;

_controller.forward(from: 0);
_controller.reverse(from: 1);
_controller.repeat();
_controller.stop();
_controller.reset();

_controller.addListener(() {
  print('AnimationController: ${_controller.value}');
});

_controller.addStatusListener(
  (status) => print('AnimationController: $status'),
);

// and many more
```

AnimationController

It can be ugly,
but its effects are beautiful.

```dart
class _AnimationControllerDemoState extends State<_AnimationControllerDemo>
    with SingleTickerProviderStateMixin {
  late final AnimationController _controller;

  @override
  void initState() {
    super.initState();
    _controller = AnimationController(
      vsync: this,
      duration: const Duration(seconds: 2),
    );

    _controller.repeat(reverse: true);
  }

  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }

  @override
  Widget build(BuildContext context) {
    return Transform.scale(
      scale: _controller.value,
      child: const Text('hello'),
    );
  }
}
```

AnimationController

It is not all about *AnimationController*
Animation is a close friend as well.

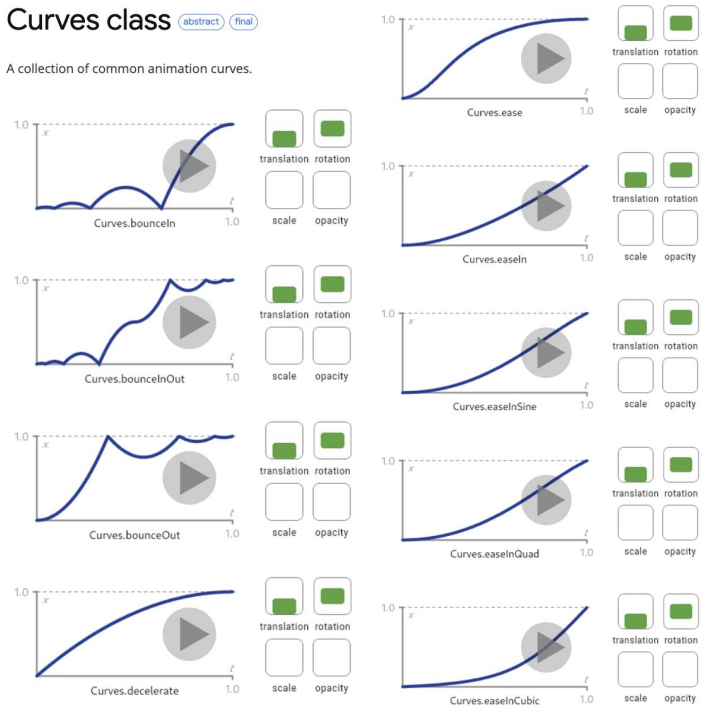
```dart
class _MyPageState extends State<MyPage>
    with SingleTickerProviderStateMixin {
  late final AnimationController _animationController;
  late final Animation<double> _animation;

  @override
  void initState() {
    super.initState();

    _animationController = AnimationController(
      vsync: this,
      duration: const Duration(seconds: 2),
    );

    _animation = CurvedAnimation(
      parent: _animationController,
      curve: Curves.elasticOut,
    );

    _animationController.repeat();
  }

  @override
  Widget build(BuildContext context) {
    return AnimatedBuilder(
       animation: _animationController,
       builder: (context, child) => Transform.scale(
       scale: _animation.value,
         child: child,
       ),
       child: ...
    );
  }
}
```

Animation curves

What is a curve?

- A function that maps **time progress (t ∈ 0..1) → value progress**
- It does **not change the animation duration**, only its **dynamics**
- The same animation with a different curve

  = a completely different **UX feel**



Animation Springs

What is a spring?

- A **physical spring model**, not a mathematical curve
- Simulates **object motion in real time**
- Reacts to:
  - animation interruption
  - target change
  - current velocity

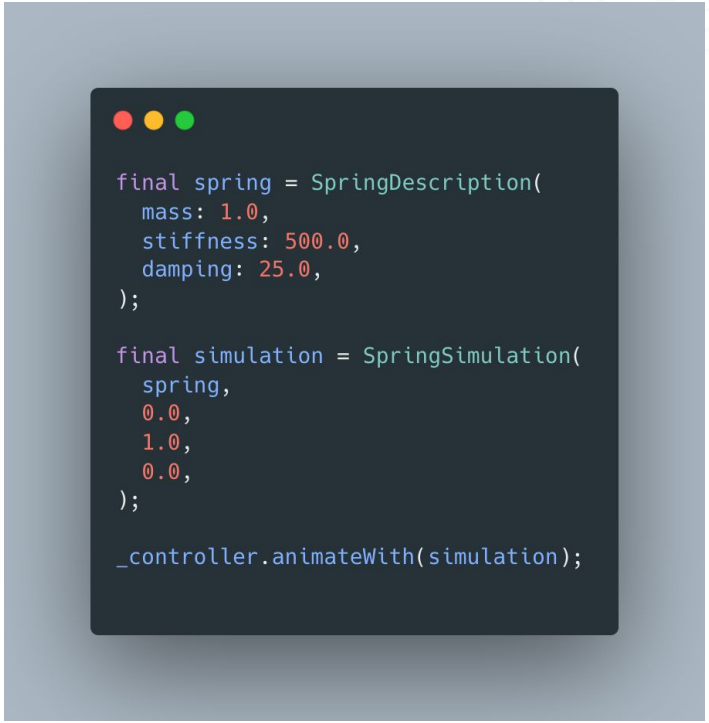
Animation Springs

Spring APIs

- *SpringDescription*
  - *mass, stiffness, damping*
- *SpringSimulation*
  - *(spring, start, end, velocity)*
- *AnimationController.animateWith(simulation)*
  - controller does not need *duration*
  - animation duration is deducted from the physics

Animation Springs

Connects to what we already know!



```
final spring = SpringDescription(
  mass: 1.0,
  stiffness: 500.0,
  damping: 25.0,
);

final simulation = SpringSimulation(
  spring,
  0.0,
  1.0,
  0.0,
);

_controller.animateWith(simulation);
```

Curves vs Springs

Curves vs Springs

Curves

- Mathematical function
- Predefined behavior
- Fixed duration
- No concept of velocity
- No reaction to interruption
- Deterministic

Typical use-cases

- fade in / out
- icon scaling
- screen transitions
- micro-interactions
- loading indicators

Springs

- Physical model (motion simulation)
- No fixed duration
- Has velocity
- Reacts to target changes
- Reacts to interruption
- Non-deterministic (input-dependent)

Typical use-cases

- drag & swipe
- bottom sheets
- cards (Tinder-style)
- fling after release

What can be animated?

Tweens

Tween

What is Tween?

- Maps values *begin -> end*
- Does not animate time - only maps the values from the controller
- Works with various types.

Tweens

Some other examples

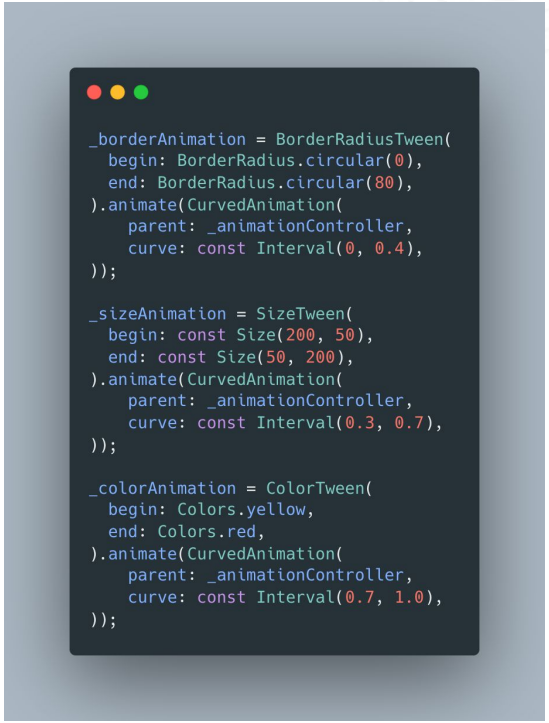
```
_animation = ColorTween(
  begin: Colors.yellow,
  end: Colors.orange,
).animate(_animationController);
```

```
_animation = SizeTween(
  begin: const Size(200, 50),
  end: const Size(50, 200),
).animate(_animationController);
```

```
_animation = BorderRadiusTween(
  begin: BorderRadius.circular(0),
  end: BorderRadius.circular(80),
).animate(_animationController);
```

Tweens

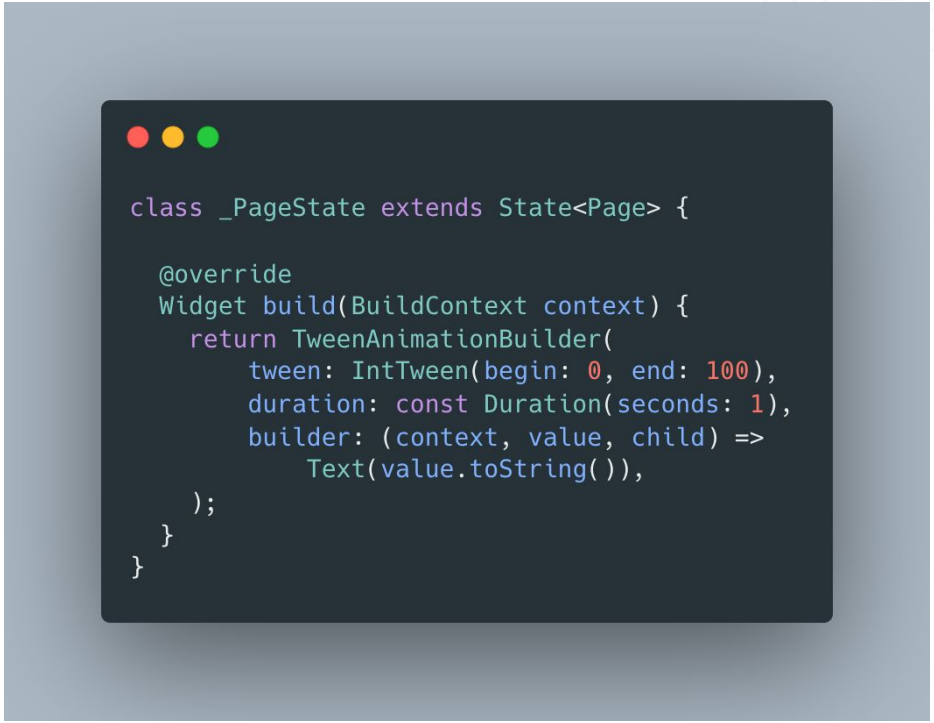
Multiple animations can be bounded to the same controller.



```
_borderAnimation = BorderRadiusTween(
  begin: BorderRadius.circular(0),
  end: BorderRadius.circular(80),
).animate(CurvedAnimation(
    parent: _animationController,
    curve: const Interval(0, 0.4),
));

_sizeAnimation = SizeTween(
  begin: const Size(200, 50),
  end: const Size(50, 200),
).animate(CurvedAnimation(
    parent: _animationController,
    curve: const Interval(0.3, 0.7),
));

_colorAnimation = ColorTween(
  begin: Colors.yellow,
  end: Colors.red,
).animate(CurvedAnimation(
    parent: _animationController,
    curve: const Interval(0.7, 1.0),
));
```

Tweens

TweenAnimationBuilder can
come in handy and
save us some boilerplate.



```
class _PageState extends State<Page> {

  @override
  Widget build(BuildContext context) {
    return TweenAnimationBuilder(
        tween: IntTween(begin: 0, end: 100),
        duration: const Duration(seconds: 1),
        builder: (context, value, child) =>
            Text(value.toString()),
    );
  }
}
```

Can we make it even easier?

of course!

Transitions

RelativePositionedTransitio

Animated version of Positioned which transitions the child's position based on the value of rect relative to a bounding box with the specified size.

RotationTransition

Animates the rotation of a widget.

ScaleTransition

Animates the scale of transformed widget.

SizeTransition

Animates its own size and clips and aligns the child.

SlideTransition

Animates the position of a widget relative to its normal position.

SliverFadeTransition

Animates the opacity of a sliver widget.

Find more widgets in the widget catalog.

Transitions

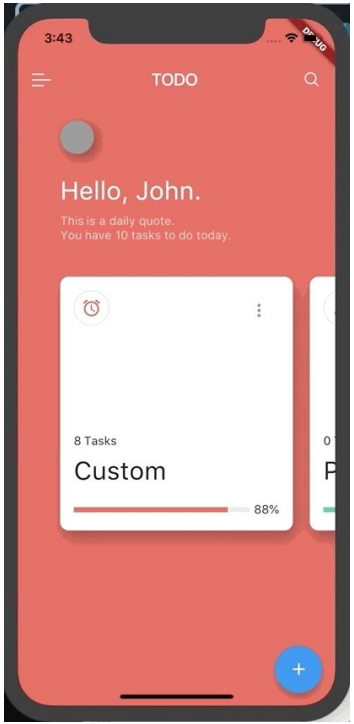
Just look for proper Transition widget :)

```
@override
Widget build(BuildContext context) {
  return Scaffold(
    body: Center(
      child: RotationTransition(
        turns: _animationController,
        child: Container(
          color: Colors.red,
          width: 300,
          height: 300,
        ),
      ),
    ),
  );
}
```

Some other tricks…

Heroes

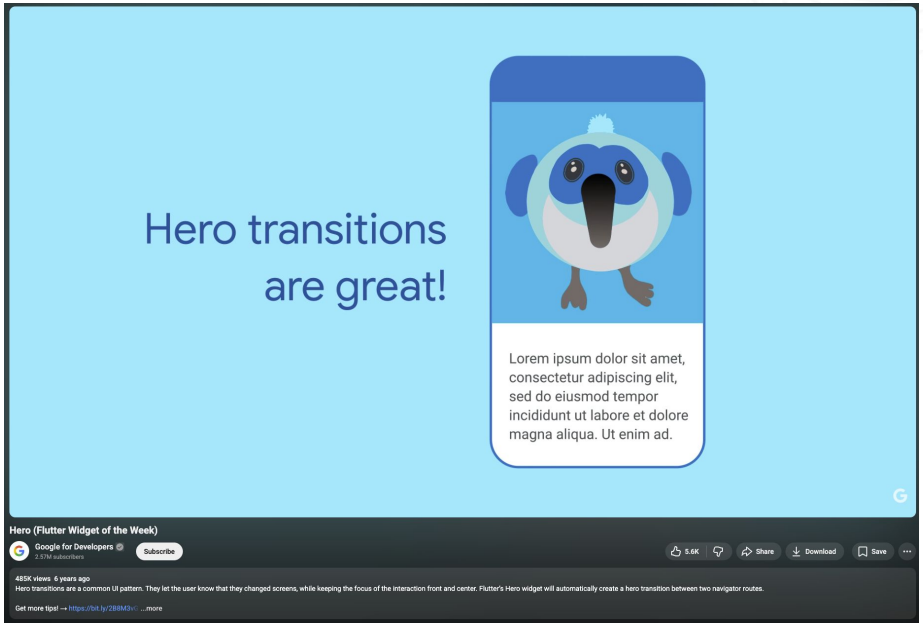
Heroes



https://github.com/flutter/flutter/issues/17627

Heroes

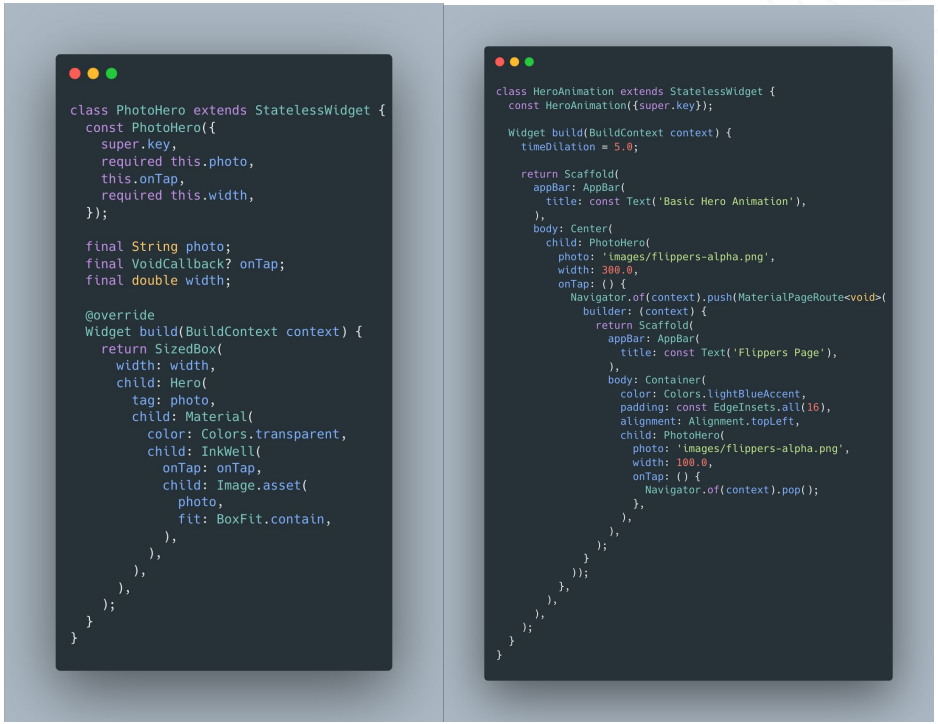
The hero refers to the widget that flies between screens



https://www.youtube.com/watch?v=Be9UH1kXFDw

Heroes

Two hero widgets with the same tag allow for good looking transition of the widget from the source route to the destination route.

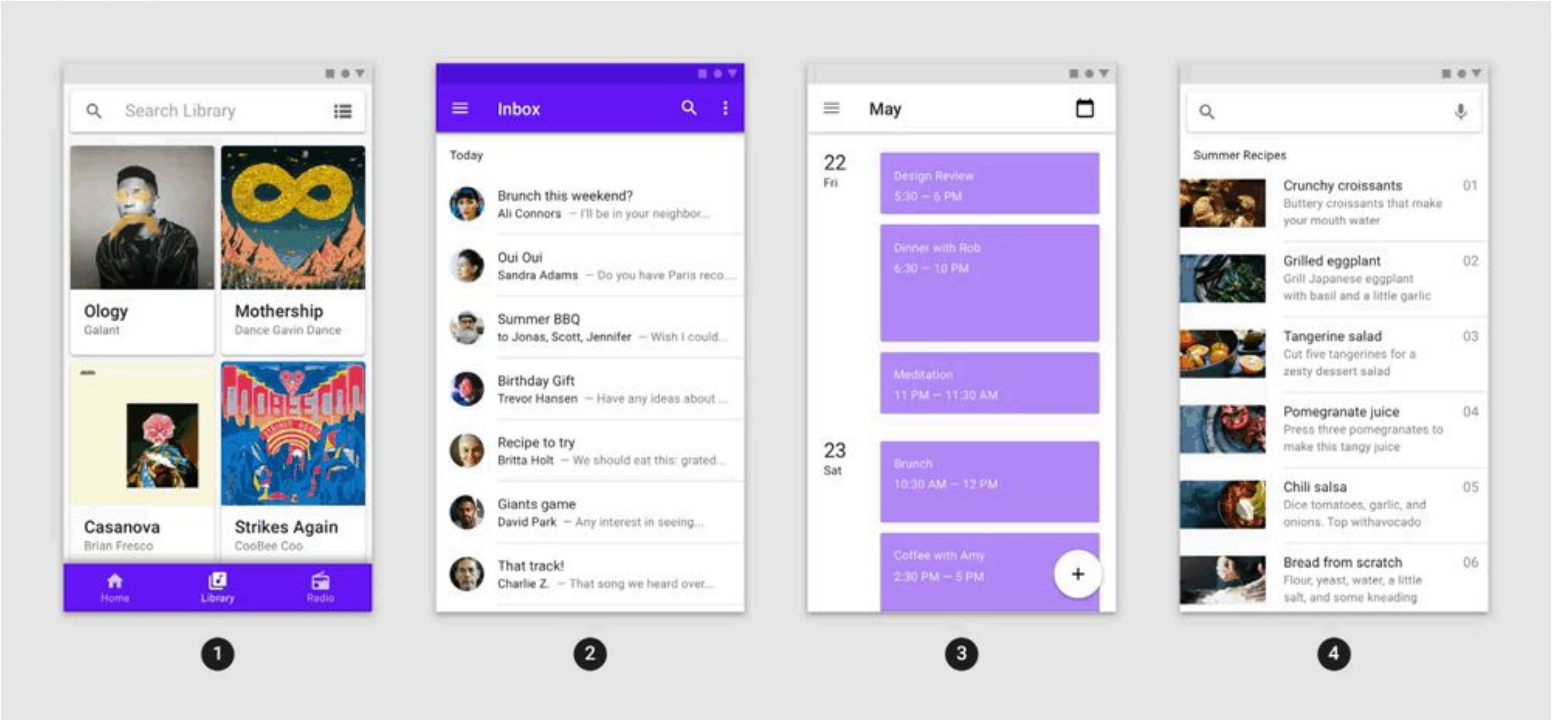


```dart
class PhotoHero extends StatelessWidget {
  const PhotoHero({
    super.key,
    required this.photo,
    this.onTap,
    required this.width,
  });

  final String photo;
  final VoidCallback? onTap;
  final double width;

  @override
  Widget build(BuildContext context) {
    return SizedBox(
      width: width,
      child: Hero(
        tag: photo,
        child: Material(
          color: Colors.transparent,
          child: InkWell(
            onTap: onTap,
            child: Image.asset(
              photo,
              fit: BoxFit.contain,
            ),
          ),
        ),
      ),
    );
  }
}
```

```dart
class HeroAnimation extends StatelessWidget {
  const HeroAnimation({super.key});

  Widget build(BuildContext context) {
    timeDilation = 5.0;

    return Scaffold(
      appBar: AppBar(
        title: const Text('Basic Hero Animation'),
      ),
      body: Center(
        child: PhotoHero(
          photo: 'images/flippers-alpha.png',
          width: 300.0,
          onTap: () {
            Navigator.of(context).push(MaterialPageRoute<void>(
              builder: (context) {
                return Scaffold(
                  appBar: AppBar(
                    title: const Text('Flippers Page'),
                  ),
                  body: Container(
                    color: Colors.lightBlueAccent,
                    padding: const EdgeInsets.all(16),
                    alignment: Alignment.topLeft,
                    child: PhotoHero(
                      photo: 'images/flippers-alpha.png',
                      width: 100.0,
                      onTap: () {
                        Navigator.of(context).pop();
                      },
                    ),
                  ),
                );
              }
            ));
          },
        ),
      ),
    );
  }
}
```

Animations package

Animations package



Animations package

High quality pre-built Animations for Flutter!

```dart
class _OpenContainer extends StatelessWidget {
  const _OpenContainer();

  @override
  Widget build(BuildContext context){
    return OpenContainer(
        transitionBuilder: Duration(miliseconds: 1000),
        closedBuilder: (_, __) => const Center(
            child: ListTile(title: Text('Item')),
        ),
        openBuilder: (_, __) => Scaffold(
            appBar: AppBar(title: const Text('Page 2')),
            body: const Center(child: Text('page2')),
        ),
    );
  }
}
```

Rive

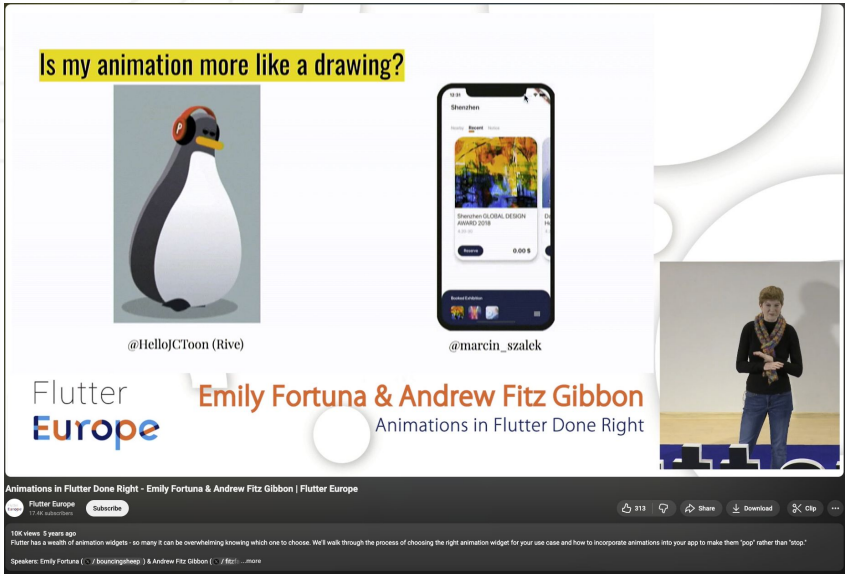
Rive

Build the animation within
the editor and just
load it in the Flutter app!

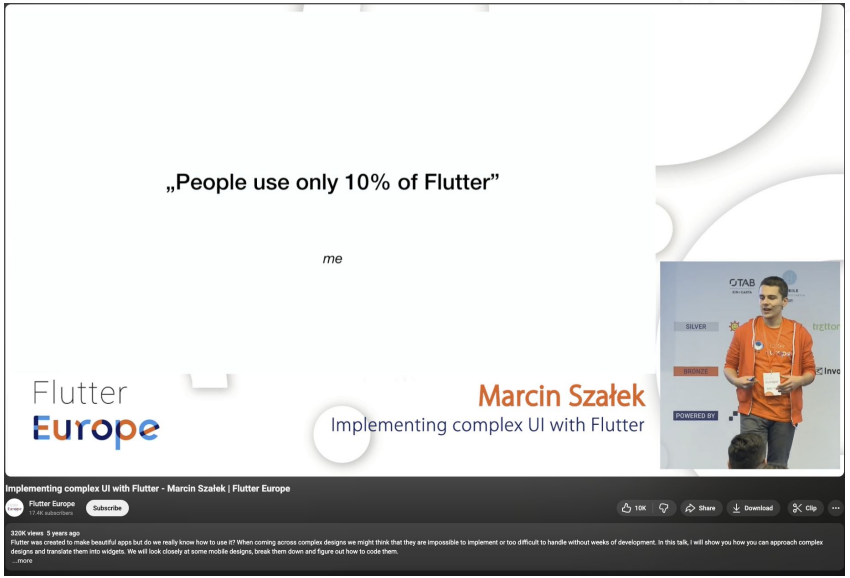
https://medium.com/@RotenKiwi/rive-for-flutter-animations-a99bfdd8f6cc

Next steps

Next steps



https://www.youtube.com/watch?v=pq791sJtWBQ



https://www.youtube.com/watch?v=FCyoHclCqc8

Next steps

https://docs.flutter.dev/ui/animations

Introduction to animations

UI › Animations

Well-designed animations make a UI feel more intuitive, contribute to the slick look and feel of a polished app, and improve the user experience. Flutter's animation support makes it easy to implement a variety of animation types. Many widgets, especially Material widgets, come with the standard motion effects defined in their design spec, but it's also possible to customize these effects.

Choosing an approach

There are different approaches you can take when creating animations in Flutter. Which approach is right for you? To help you decide, check out the video, How to choose which Flutter Animation Widget is right for you? (Also published as a *companion article*.)